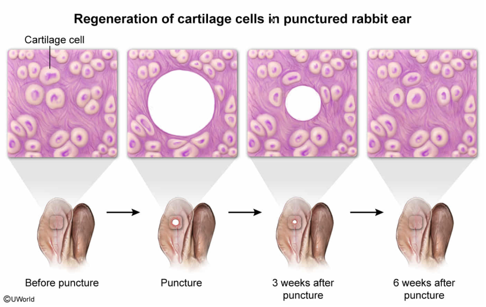
Tissue regeneration has an important role in which of the following circumstances?

- A. Generating a multicellular embryo from a zebrafish zygote [1%]
- B. Producing neurons from chicken embryonic stem cells [2%]
- ✔ C. Producing new cartilage cells in a punctured rabbit ear [38%]
- D. Creating scar tissue in human skin that has been injured [57%]

Correct
39% answered correctly

Explanation:



Regeneration is the **regrowth** of the *same* functional tissues or organs previously present in an organism after loss due to injury. For example, the human liver can regenerate after it has been damaged. The capacity for regeneration varies among species.

In the image above, a rabbit ear is punctured (ie, injured) and regeneration causes the *regrowth* of the same functional ear tissue (ie, tissue such as cartilage that was present prior to the injury). Therefore, **regeneration** has an important role in **producing new cartilage cells** in a punctured rabbit ear.

(Choice A) During development, mitosis results in the growth of an organism from a single-cell stage (eg, zebrafish zygote) to a multicellular stage (ie, embryo). During the development of an embryo from a zygote, *new* cells and tissues are created via cell division. However, because these new cells are *not* replacing functional tissue that was *previously present* in the organism, this is not an example of regeneration.

(Choice B) The process of differentiation creates tissue-specific cells (eg, neurons) from stem cells (eg, undifferentiated chicken embryonic stem cells). During differentiation, cells gain tissue-specific functions; however, these cells are *not* regrown (ie, *replaced*) after their loss. Therefore, differentiation *differs* from regeneration.

(Choice D) After an injury, scar tissue often replaces tissue that was present before the injury (eg, human skin tissue). Because scar tissue is a *different* type of tissue than what was present before the injury (epithelial tissue), this is *not* an example of regeneration.

Educational objective:
Regeneration is the regrowth of tissues or organs previously present in an organism after loss due to injury. These regrown cells function in the same way as the original cells.

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